IMPLEMENTATION

System Portal

Architecture, implementation details, and every technology/tool used in this build.

1. Overview

System Portal is a static, multi-page gamified progression dashboard inspired by Solo Leveling's 'System' UI. It runs entirely in the browser (no backend). Users must authenticate before accessing the site; each account gets its own local progression profile (quests, stats, XP, gold, streaks, check-ins, transformation posts). All 270 supplied frame images are used for autoplaying cinematic chapter/hero sequences.

2. Repository & file structure

Path	Purpose
`index.html + 10 pages`	Home, 4 chapter pages (awakening, extraction, arise, army), 6 dashboard modules (status, quests, vision, advisor, progress, inventory). All protected by auth guard.
`login.html`	Sign-in/sign-up screen with Google Identity Services button (when configured) and local email/password auth, plus animated visuals.
`styles/design-system.css`	Global tokens, reset, typography, layout, components, navigation, user menu, empty/history/check-in states, responsive rules.
`styles/pages.css`	Page composition: hero/chapter stages, HUD, modules, gauges, quests, chat, inventory, modals, toasts, motion/FX, boot sequence, ticker, notifications.
`styles/auth.css`	Login layout and animations (particles canvas, orbs, rings, scan beam, entrance transitions).
`scripts/config.js`	Runtime config: app name, Google OAuth client ID placeholder, session hours, built-in admin email/name/password.
`scripts/auth.js`	Client-side auth: session storage/expiry, route guard, local account hashing (PBKDF2-SHA-256), sign-in/up, admin seeding, Google credential decode + audience/expiry checks.
`scripts/login.js`	Login UI controller: tabs, form handling, GIS initialization/rendering, particle animation on the visual panel.
`scripts/app.js`	Shared app: shell/header/footer, per-user Store (v3), progression math, frame player, FX, and page initializers for every module.
`ezgif-frame-001.jpg … 270.jpg`	270 animation frames used by the canvas frame player (home slice + full chapter slices).
`generate_build_report.py`	ReportLab script that generates the 3-page build report PDF (SYSTEM_PORTAL_BUILD_REPORT.pdf).
`generate_implementation_pdf.py`	ReportLab script that generates this implementation PDF (IMPLEMENTATION.pdf).
`verify.py`	Validation helper for pages/structure/IDs (project checks).

3. Authentication & access control

- **Route guard:** every protected HTML page loads `config.js`, `auth.js`, then calls `SystemAuth.requireAuth()` in `<head>` to redirect unauthenticated visitors to `login.html?next=…` before content paints.
- **Local accounts:** email/password accounts stored in `localStorage` (`system_portal_users_v1`). Passwords are never stored in plain text: PBKDF2-SHA-256 via Web Crypto with 210,000 iterations and a random 16-byte salt per account.
- **Built-in admin:** configured in `scripts/config.js` (admin@system.local / AriseAdmin#2026 by default) and ensured/hashed on admin sign-in.
- **Google Sign-In:** Google Identity Services (`accounts.google.com/gsi/client`) renders a real sign-in button when a valid OAuth Web client ID is set. The ID token payload is decoded in-browser; audience and expiry are checked. **Limitation (static site):** token signature cannot be server-verified without a backend.
- **Sessions:** stored in `localStorage` (`system_portal_session_v1`) with configurable expiry (default 12 hours); sign-out and cross-tab storage events clear/redirect sessions.

4. Data model & progression (real, derived)

- **Per-user storage:** `system_portal_state_v3_<email>`. New accounts start at Level 1, Rank E, 0 XP/gold/AP, stats 10/10/10/10/10, no preloaded quests/items/check-ins.
- **XP & leveling:** XP to next level = `100 + (level-1)*50`. Completing quests grants XP; level-ups grant +3 ability points. Ranks/titles are derived from level (E→S).
- **Quests:** user-created with title, rank (E–S multiplier), and duration. Rewards computed as duration × rank multiplier (XP/gold/AP shown before issuing). Deadlines are real wall-clock timestamps; expiry applies fatigue/penalty rules; history is recorded.
- **Stats & derived vitals:** STR/AGI/VIT/INT/PER allocated from earned AP. Max HP/MP and gauges are derived from level and stats. Streaks computed from completed quest dates.
- **Vision/check-ins:** real camera capture via `getUserMedia`, frames downscaled to canvas and stored locally as JPEG data URLs (no upload, no fake posture analysis). Reminders use real due timestamps.
- **Progress feed:** user-uploaded before/current photos resized client-side and stored locally with consent; no fabricated sample posts.
- **Inventory/milestones:** records and skills unlock from real counts/levels/streaks (no stat buffs). Advisor answers from the user's actual state via deterministic rules.

5. Animation & visual system

- **Frame player:** custom canvas player preloads frame slices, draws with DPR scaling and cover-fit (bottom watermark cropped), autoplays at fixed fps, loops, pauses offscreen (IntersectionObserver), supports play/pause + progress UI.
- **Sequences:** Home hero plays frames 1–90 @20fps with boot/loading overlay; chapters autoplay their slices @15fps (Awakening 1–65, Extraction 66–130, Arise 131–190, Army 191–270).
- **FX & motion:** ambient particle canvas, scanlines/vignette, HUD corners, tickers, notification panels, scroll reveals, counters, reading progress, hero signals, and login-page animations (canvas particles, orbs, rings, scan beam, entrance transitions). Honors `prefers-reduced-motion`.
- **Brand/UI:** custom SVG monolith logo mark, "SYSTEM / PORTAL" wordmark, dark Solo Leveling 'System' palette (violet/cyan on near-black) with Antigravity-inspired layout primitives.

6. Technologies & tools used

Category	Used for	Details
HTML5	Structure	Semantic multi-page documents, forms, canvas/video elements, metadata.
CSS3	Styling & motion	Custom properties, Grid/Flexbox, gradients, filters, backdrop-filter, keyframe animations, responsive breakpoints, reduced-motion media queries.
JavaScript (ES5/ES6+)	Application logic	Vanilla JS modules (IIFEs), DOM APIs, Canvas 2D, Intersection/ResizeObserver, FileReader, URL/createObjectURL, timers, async/await for crypto flows.
Web Crypto API	Password security	PBKDF2-SHA-256 (210k iterations), random salts, subtle.importKey/deriveBits for local account hashing.
localStorage / sessionStorage	Persistence	Per-user state, accounts, sessions, boot flag; cross-tab session sync via storage events.
Google Identity Services	Google Sign-In	GSI client script, ID token credential callback, client-side payload decode/audience/expiry checks (client ID required).
MediaDevices API	Camera check-ins	getUserMedia for live preview and snapshot capture to canvas.
Notifications API	Reminders	Optional browser notifications when permission is granted.
Python 3	Tooling	Local static server (`python -m http.server 8090`) and PDF/report generation scripts.
ReportLab	PDF generation	Programmatic creation of SYSTEM_PORTAL_BUILD_REPORT.pdf and IMPLEMENTATION.pdf.
Git & GitHub CLI (gh)	Versioning & publish	Repository initialization, commits, and publishing the public `system` repository to GitHub.
Assets	Animation frames	270 JPEG frames (ezgif-frame-001..270.jpg) rendered through the canvas player.

7. How to run locally

- From the project folder run: `python -m http.server 8090`
- Open `http://localhost:8090/login.html` and sign in (admin: admin@system.local / AriseAdmin#2026, or create a local account, or configure Google client ID in `scripts/config.js`).
- Data is stored in the browser's localStorage for the signed-in email on that device/origin.

8. Notes & limitations

- No backend/database: accounts, sessions, and progression live in the browser; clearing site data resets them.
- Google token signatures are not server-verified in this static build (audience/expiry checked client-side only).
- Admin credentials are defined in client-side config for this local/demo setup; treat accordingly for any shared deployment.
- Fan-made productivity prototype. Solo Leveling is owned by Chugong & Redice Studio.

Generated for the System Portal public repository.